Chapter 17 Raw Textbook Content

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Textbook of Minimally Invasive Spine Surgery, Concepts and Surgical Techniques

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1. History and Progress of Minimally Invasive Spine Surgery

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1.1 Introduction

Surgical management of spinal disorders has a rich and evolving history that spans centuries. Early documentation of spinal injuries can be traced back to ancient Egypt, where the Edwin Smith Surgical Papyrus described several cases of vertebral trauma and their treatments [1–4]. In ancient Greece, Hippocrates advanced the understanding of spinal anatomy and pathology, pioneering techniques such as spinal manipulation and traction [5–7]. Over the next several centuries, further contributions by Galen and other scholars refined these ideas, introducing the concept of spinal alignment and stabilization [8–11]. By the nineteenth century, significant progress was made in surgical practice, culminating in Alban Smith's performance of the first laminectomy in the United States in 1829 [12–14]. The introduction of aseptic techniques and anesthesia soon after revolutionized the safety and scope of spinal surgery, leading to widespread adoption of open surgical approaches to treat a range of spinal conditions [15–18].

Traditional open spine surgery has been the cornerstone of managing spinal pathologies. These approaches, while effective, relied on wide exposure through large incisions, necessitating significant muscle dissection and soft tissue disruption to achieve the desired surgical objectives. Although they provided excellent visualization and access, the collateral damage associated with such techniques often resulted in longer recovery times, higher complication rates, and greater postoperative morbidity [19–21]. These factors became increasingly important as spine surgery expanded to treat more complex conditions in a growing patient